

forecasting injury risk. The logic is straightforward: track short-term versus long-term load and avoid 'spikes'.

But here's the problem — these ratios don't measure cause, only correlation.

They fail to account for muscle- and tissue-specific adaptation and specific individual context. Simulations have shown that even random data can produce 'predictive' relationships with injury using ACWR. Arbitrary cut-offs (like the 'sweet spot' of 0.8–1.3) have no biological basis.

The simplicity is somewhat why these ratios and sweet spots gained popularity and acceptance. However, they are an overly simplified explanation that ignores the bigger picture, social contexts, or complex variables. It is a reductionist point of view that fails to account for the complexity of training adaptations and injury.

In short, many of the metrics we've leaned on are statistical illusions, not actionable science.

WHAT CAN TL MONITORING DO?

Here's where it gleams when TL is used properly as a valuable tool:

- ✦ Verifying programme and plan adherence — Did the athlete actually do what was planned?

- ✦ Tracking tolerance over time — Is the athlete handling progressive overload as anticipated?

- ✦ Identifying large digressions — Is an athlete under- or over-training compared to expectations?

- ✦ Appraising adjustments — In conjunction with subjective and objective responses.

TL is like a car dashboard gauge. It tells you what's happening currently, but it doesn't drive the car and cannot predict a crash. It's a tool to be used in varied action and reaction potential.

CLINICAL REASONING & LAYERED DECISION-MAKING

Injury prevention is not a formula; it's a multifarious, interdisciplinary process with protocols in place. Success comes from coalescing varied aspects of the process through bespoke protocols.

- ◆ Evidence-based planning: Use training principles like overload, progression, and recovery timelines.

- ◆ Athlete-centred monitoring: Include subjective metrics (e.g., soreness, fatigue), objective tests (e.g., jump performance), and apt feedback mechanisms.

- ◆ Collective decision-making: Coaches, physios, other support staff, and athletes should collectively make informed, individualised decisions.

- ◆ Attention to tissue-specific load tolerance: Rather than chasing arbitrary numbers, build capacity across body structures.

THE RISKS OF OVER-RELIANCE ON TL METRICS

Here's what happens when we oversimplify complex systems:

- ✦ We ignore the multifactorial nature of injuries (e.g., biomechanics, prior injury, psychosocial stressors).

- ✦ We create false narratives with athletes, coaches, and organisations.

- ✦ We jeopardise missing the real warning signs in favour of chasing 'red zones' on a dashboard.

- ✦ Worst of all, we may undermine the trust of athletes, staff, and the front office by promising control over something inherently uncertain.

HOW TO USE AND NOT USE TRAINING LOAD

Let's be honest: we can't eliminate injury risk. But we can reduce it through intelligent, adaptive planning. Training load data can be astoundingly useful when applied properly.

Use TL to:

- ◆ Track adherence to training plans.

- ◆ Inform decisions alongside other metrics like movement quality, readiness, and recovery data.

- ◆ Adjust load based on athlete feedback and observed tolerance over time.

- ◆ Support return-to-play readiness by monitoring gradual progression and identifying deviations.

Don't use TL to:

- ◆ Predict injuries with precision.

- ◆ Presuppose control over complex systems.

- ◆ Replace judgement with algorithms.

- ◆ Distort communication with stakeholders.

The bottom line is that training load is a support tool, not a silver bullet. It works best when used in perspective, combined with clinical deduction, and always percolated through the lens of the coach's experience.

